

INTERMEDIATE QUANTS – 3

Directions for question 1: Type in your answer in the input box provided in the question.

The n^{th} term of a sequence, a_n , is defined as $a_n =$

$$\frac{n+1}{2}$$

when n is odd

$$= \frac{n}{2}$$

when n is even

$$\sum_{n=1}^{80} a_n =$$

Solution:

$a_{2k-1} = a_{2k} = k$, where k is any positive integer.

$$\sum_{n=1}^{80} a_n = (a_1 + a_2) + (a_3 + a_4) + (a_5 + a_6) + \dots + (a_{79} + a_{80})$$

$$= (1 + 1) + (2 + 2) + (3 + 3) + \dots + (40 + 40)$$

$$= 2(1 + 2 + 3 + \dots + 40) = 2 \left(\frac{(40)(41)}{2} \right) = 1640$$

Ans : (1640)

Directions for questions 2 to 5: Select the correct alternative from the given choices.

How many solutions does $x^2 + 9$

$|x|$

$+ 20 = 0$ have?

☒ a) 4 (Your answer - ✖)

☐ b) 0

☐ c) 2

☐ d) 3

Solution:

$$x^2 + 9|x| + 20 = 0$$

$$|x|^2 + 9|x| + 20 = 0$$

$$(|x| + 4)(|x| + 5) = 0$$

$$|x| = -4 \text{ or } -5$$

But $|x|$ cannot be negative. $\therefore$ No solution exists.

Choice (B)

Directions for questions 2 to 5: Select the correct alternative from the given choices.

2 machines, 4 men and 32 women can complete a piece of work in 6 days. 27 men and 96 women can complete the work in 2 days. The ratio of the rates of work of each machine and each man is _____

- ☐ a) 3 : 2
- ☐ b) 5 : 2
- ☐ c) 2 : 1
- ☐ d) 3 : 1

Solution:

Let the rates of work of each machine, each man and each women (in units/day) be s , m and n respectively.

Work that can be completed by 2 machines, 4 men, 32 women in a day = $2s + 4m + 32n$ and hence in 6 days = $6(2s + 4m + 32n)$

Work = $6(2s + 4m + 32n)$

Also, work = $2(27m + 96n)$

Work = $6(2s + 4m + 32n) = 2(27m + 96n)$

$12s + 24m + 192n = 54m + 192n$

$12s = 30m$

$$\frac{s}{m} = \frac{5}{2}$$

Choice (B)

Directions for questions 2 to 5: Select the correct alternative from the given choices.

60 is divided into four parts which are proportional to 3, 5, 7 and y . If the smallest part is $1\frac{1}{2}$

$\frac{1}{2}$

, and $y < 100$, find y .

- ☐ a) 99
- ☐ b) $\frac{13}{5}$
- ☐ c) $\frac{5}{13}$
- ☐ d) Data inconsistent

Solution:

$$\text{Either } \frac{3}{3+5+7+y} (60) = \frac{3}{2} \text{ or}$$

$$\frac{y}{3+5+7+y} (60) = \frac{3}{2}$$

From the first equation, we get $y = 105$ and from the second equation, $y = \frac{15}{39} = \frac{5}{13}$.

As $y < 100$, $y = \frac{5}{13}$.

Choice (C)

Directions for questions 2 to 5: Select the correct alternative from the given choices.

PQR is a triangle. A, B and C are points on PQ, QR and RP respectively. $PA : AQ = QB : BR = RC : CP = 2 : 3$. Find the ratio of the areas of the triangles ABC and PQR.

- ☐ a) 9 : 25
- ☐ b) 11 : 25
- ☐ c) 7 : 25
- ☐ d) Cannot be determined

Solution:

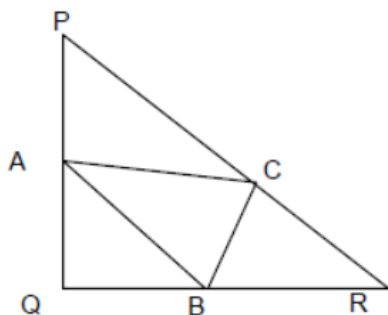
Let the areas of PAC, QBA, RCB and PQR be X, Y, Z, Δ respectively.

$$\frac{X}{\Delta} = \frac{PA}{PQ} \cdot \frac{PC}{PR} = \frac{2}{5} \times \frac{3}{5} = \frac{6}{25}.$$

$$\text{Similarly } \frac{Y}{\Delta} = \frac{Z}{\Delta} = \frac{6}{25}$$

$$\text{Area of ABC} = \Delta - (\text{sum of areas of PAC, QAB, RBC}) = \left(1 - \frac{18}{25}\right) \Delta = \frac{7}{25} \Delta$$

Ratio of the areas of ABC and PQR is 7 : 25.



Choice (C)

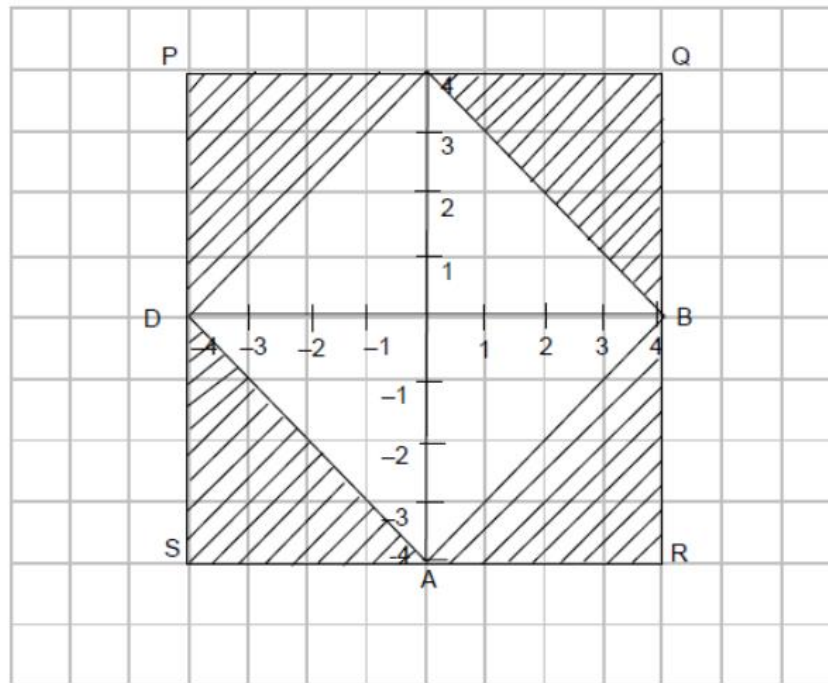
Directions for question 6: Type in your answer in the input box provided below the question.

Find the area bounded by $|x| = 4$, $|y| = 4$, and

$$|x| + |y|$$

$$= 4.$$

Solution:



Area bounded by $|x| + |y| = 4$, $|x| = 4$, $|y| = 4$ is sum of the areas of ΔPCD , ΔCQB , ΔBAR and ΔDSA . (the shaded region).

$ABCD$ is a square with side $4\sqrt{2}$. While $PQRS$ is a square of side 8.

Required area = $8^2 - (4\sqrt{2})^2 = 64 - 32 = 32$.

Ans : (32)

Directions for questions 7 to 12: Select the correct alternative from the given choices.

The operator # is defined as follows:

$$p \# q = p + q - pq$$

Consider the following statements:

I: # is commutative

II: # is associative

Which of the statement is/are true?

- ☐ a) Only I
- ☐ b) Only II
- ☐ c) Both I and II
- ☐ d) Neither I nor II

Solution:

If # is commutative, $a \# b = b \# a$

$$p \# q = p + q - pq = q + p - qp = q \# p$$

$\therefore$ # is commutative.

If # is associative, $(p \# q) \# r = p \# (q \# r)$

$$(p \# q) \# r = (p + q - pq) \# r$$

$$= p + q - pq + r - (p + q - pq)r$$

$$= p + q + r - pq - pr - qr + pqr$$

$$p \# (q \# r) = p \# (q + r - qr)$$

$$= p + q + r - qr - p(q + r - qr)$$

$$= p + q + r - qr - pq - pr + pqr$$

$$(p \# q) \# r = p \# (q \# r)$$

is associative.

is commutative as well as associative.

Choice (C)

Directions for questions 7 to 12: Select the correct alternative from the given choices.

If θ is an acute angle satisfying $(\cos\theta)(\cot\theta) = \sin\theta$ and $m\operatorname{cosec}^2\theta + \sec^2\theta = 15$, the value of m is _____.

- ☐ a) $\frac{7}{2}$
- ☐ b) $\frac{13}{2}$
- ☐ c) $\frac{19}{2}$
- ☐ d) $\frac{5}{2}$

Solution:

$$(\cos\theta)(\cot\theta) = \sin\theta$$

$$\Rightarrow (\cos\theta) \left(\frac{\cos\theta}{\sin\theta} \right) = \sin\theta$$

$$\cos^2\theta = \sin^2\theta$$

$$\sin^2\theta + \cos^2\theta = 1, \therefore \sin^2\theta = \cos^2\theta = \frac{1}{2}$$

$$m\operatorname{cosec}^2\theta + \sec^2\theta = 15$$

$$m(2) + 2 = 15 \Rightarrow m = \frac{13}{2}.$$

Choice (B)

Directions for questions 7 to 12: Select the correct alternative from the given choices.

If $f(x - 1) + f(x + 1) =$

$$\sqrt{2}$$

$f(x)$, $f(1) =$ _____.

- ☐ a) $f(5)$
- ☐ b) $-f(5)$
- ☐ c) $\frac{f(5)}{2}$
- ☐ d) $-2f(5)$

Solution:

$$f(x-1) + f(x+1) = \sqrt{2} f(x) \rightarrow (1)$$

$$\text{Substituting } x = 2 \text{ in (1): } f(1) + f(3) = \sqrt{2} f(2) \rightarrow (2)$$

$$\text{Substituting } x = 3 \text{ in (1): } f(2) + f(4) = \sqrt{2} f(3) \rightarrow (3)$$

$$\begin{aligned}\text{Substituting } x = 4 \text{ in (1): } f(3) + f(5) &= \sqrt{2} f(4) \\ &= \sqrt{2} (\sqrt{2} f(3) - f(2)) \quad (\because \text{From (3)}) \\ &= 2f(3) - \sqrt{2} f(2) \rightarrow (4)\end{aligned}$$

$$\begin{aligned}(1) + (4): f(1) + f(5) &= 0 \\ f(1) &= -f(5)\end{aligned}$$

Choice (B)

Directions for questions 7 to 12: Select the correct alternative from the given choices.

Three terms are in harmonic progression and the sum of their reciprocals is 30. If the ratio of the second and the third terms is 3 : 2, find the smallest among the three.

- ☐ a) $\frac{1}{5}$
- ☐ b) $\frac{1}{10}$
- ☐ c) $\frac{1}{15}$
- ☐ d) None of these

Solution:

The reciprocals of the terms which are in HP, are in AP.

$\therefore$ Let the reciprocals of the terms be

$$a-d, a, a+d$$

$$\therefore 3a = 30$$

$$\Rightarrow a = 10 \text{ and}$$

$$\frac{a+d}{a} = \frac{3}{2} \therefore \left(\frac{1}{a} \text{ and } \frac{1}{a+d} \text{ are in the ratio } 3:2 \right)$$

$$\Rightarrow 2a + 2d = 3a$$

$$\Rightarrow d = 5$$

$$\therefore \text{The terms are } \frac{1}{5}, \frac{1}{10}, \frac{1}{15}.$$

Choice (C)

Directions for questions 7 to 12: Select the correct alternative from the given choices.

Which of the following can be concluded about $\log_2 34$?

☐ a) It is rational and more than 5.5.

☐ b) It is irrational and more than 5.5.

☒ c) It is rational and less than 5.5. (Your answer - ✕)

☐ d) It is irrational and less than 5.5.

Solution:

Let $k = \log_2 34 \therefore 2^k = 34$

Suppose k is rational. Let $k = \frac{p}{q}$, where p and q are integers.

$$2^{p/q} = 34$$

$$2^p = 34^q$$

L.H.S is a power of 2 while R.H.S is not.

$\therefore$ L.H.S $\neq$ R.H.S.

$\therefore$ The assumption we started with was wrong i.e., k is irrational.

$$2^{5.5} = 2^5(0.5) = 32\sqrt{2} \text{ which is more than } 32(1.4) = 44.8$$

$$2^{5.5} > 44.8 > 34 \text{ i.e., } 2^{5.5} > 34$$

$$5.5 > \log_2 34 \text{ i.e., } \log_2 34 < 5.5 \text{ i.e., } k < 5.5.$$

$k < 5.5$ and k is irrational.

Choice (D)

Directions for questions 7 to 12: Select the correct alternative from the given choices.

The lines $x + y - 9 = 0$ and $x + y + 5.5 = 0$ are tangents to a circle. Find the radius of the circle.

☐ a) $\frac{35}{4\sqrt{3}}$

☐ b) $\frac{29}{4\sqrt{2}}$

☐ c) $\frac{39}{4\sqrt{2}}$

☐ d) $\frac{43}{4\sqrt{2}}$

Solution:

The given lines are parallel and are also tangents to the circle.

∴ The distance between the lines is the diameter of the circle

Distance between the parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\frac{|c_2 - c_1|}{\sqrt{a^2 + b^2}}$.

Distance between the given lines (D)

$$= \frac{|5.5 - (-9)|}{\sqrt{1^2 + 1^2}} = \frac{14.5}{\sqrt{2}}$$

$$\text{Radius} = \frac{14.5}{2\sqrt{2}} = \frac{29}{4\sqrt{2}}$$

Choice (B)

Directions for question 13: Type in your answer in the input box provided below the question.

The difference between the simple interest and the compound interest on a sum of money at 20% p.a, interest being compounded annually for 2 years, is Rs.132. Find the sum (in Rs.).

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

Let the sum be ₹p

Difference between simple interest and compound interest on p at 20% p.a for 2

$$\text{years} = p \left(\frac{20}{100} \right)^2$$

$$p \left(\frac{20}{100} \right)^2 = 132 \Rightarrow p = 3300$$

Ans : (3300)

Directions for question 14: Select the correct alternative from the given choices.

P starts from town X and goes towards town Y. Q starts simultaneously from Y and goes towards X. After their meeting, P takes 8 hours to reach Y and Q takes 18 hours to reach X. If the speed of P is 24 kmph, the speed of Q is _____ kmph.

- ☐ a) 36
- ☐ b) 16
- ☐ c) 30
- ☐ d) 18

Solution:

Let the speed of Q be q kmph.

Let the time taken for the two persons to meet be t hours.

Let M be the meeting point.

XM = Distance covered by P before meeting

= Distance covered by Q after meeting.

MY = Distance covered by Q before meeting

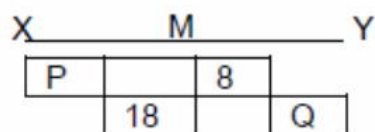
= Distance covered by P after meeting.

$$XM = 24t = (q)(18)$$

$$MY = qt = (24)(8)$$

$$\frac{XM}{MY} = \frac{24t}{qt} = \frac{(q)(18)}{(24)(8)}$$

$$q^2 = 256 \Rightarrow q = 16.$$



Choice (B)

Directions for question 15: Type in your answer in the input box provided in the question.

If x, y, z are positive numbers satisfying $x =$

$$\frac{1}{z^y}$$

$$, z =$$

$$\frac{1}{y^x}$$

$$, y =$$

$$\frac{1}{x^z}$$

the value of xyz is

Your Answer: 1 ✓

Solution:

$$x = z^{\frac{1}{y}}, z = y^{\frac{1}{x}}, y = x^{\frac{1}{z}}$$

$$\therefore x^y = z, z^x = y, y^z = x$$

$$x^y = z$$

$$\therefore z^x = (x^y)^x = x^{xy}. \text{ This is also equal to } y.$$

$$y^z = (x^{xy})^z = x^{xyz}. \text{ This is also equal to } x.$$

$$x^{xyz} = x$$

We have an equation of the form $a^m = a^n$

If we have any equation of the form $a^m = a^n$, then either

$a = -1, 0$ or 1 or else $m = n$.

$$x^{xyz} = x^1 \rightarrow (1)$$

As x is positive, x cannot be 0 or -1 .

If $x = 1$, then $z = 1, y = 1$ and $xyz = 1$

If $x \neq 1$, then indices on L.H.S and R.H.S of (1)

are equal i.e., $xyz = 1$

In either case, $xyz = 1$

Ans : (1)